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Roll No.....

M.Tech- AI IIth SEMESTER

MID –SEMESTER EXAMINATION, Feb-Mar 2025

Course Code- CPAIC03

Course Title- Soft Computing

Time-1.5 Hours

Max. Marks-15

Note- Attempt all the five questions. Missing data/information if any, maybe suitably assumed & mentioned in the answer.

Q.no	Questions	Marks	CO
		1	CO1
Q1a.	Justify the need of generalisation of crisp set as a fuzzy set.	2	CO1
Q1b.	Compare hard computing and soft computing.	1	CO1
Q2a.	Explain the components of soft computing and their importance in complex problem solving.	2	CO1
Q2b.	Consider a fuzzy set A defined on the interval $X = \{0, 1, 2, \dots, 10\}$ with the membership function: $\mu_A(x) = x/(x + 3)$ Compute α -cut for $\alpha = 0.6$.	1	CO2
Q3a.	Mention the property of a fuzzy set which is different from a crisp set.	2	CO2
Q3b.	Consider the membership function $\mu_B(x)$ $\mu_B(x) = \begin{cases} 0, & \text{if } \text{age}(x) \leq 10 \\ \frac{x-10}{15}, & \text{if } 10 \leq \text{age}(x) \leq 25 \\ 1, & \text{if } 25 \leq \text{age}(x) \leq 50 \\ \frac{80-x}{30}, & \text{if } 50 \leq \text{age}(x) \leq 80 \\ 0, & \text{if } \text{age}(x) \geq 80 \end{cases}$ Find $\mu_{\bar{B}}(x)$	1	CO2
Q4a.	Compare Fuzzy sets and Probability theory.	2	CO2
Q4b.	Explain Crossover Points and symmetric Fuzzy Set.	1	CO3
Q5a.	Explain the role of Soft Computing in Artificial Intelligence.	2	CO3
Q5b.	Following are some statements- a. Water is Hot b. Sheena is very short. c. Chinese are not very tall. d. Mostly Indians are of brown complexion. e. Generally, 5'8" above are called tall. Define fuzzy set which can be used to represent the list of above statement.		